综合练习

人员

王承周、刘奕辰、阮文璋 到课

上周作业检查

上上周作业链接: https://vjudge.net/contest/803886

开始时间 : 2026-04-12 08:30 CST

☆ 2026-0412 六队上课 (综合练习)

结束时间 : 2026-11-06 16:30 CST

已开始 : 14:09:14:04 进行中 **还剩余 :** 193:22:45:55

概览 题目 提交状态 排名 (14:09:14:00) 讨论 设置 克隆 更新 删除

Rank	Team	Score	Penalty	A 9 / 29	B 9 / 17	C 7 / 14	D 5 / 7	E 6 / 12
1	☆ yyycj	5	45364	5:13:25:22	5:13:43:52 (-1)	6:15:39:16	7:00:44:08 (-1)	6:15:52:06
2	☆ r111	5	49093	6:07:49:00 (-4)	4:14:00:00	7:06:18:00	7:08:30:00 (-1)	8:11:56:00
3	☆ wcz123bc	5	58533	7:06:48:00 (-5)	6:08:43:33	7:06:26:00	13:03:29:00	6:10:47:00 (-5)
4	☆ lyc12345bc (刘奕辰)	5	68264	7:00:39:02 (-1)	7:01:10:00 (-2)	7:01:46:30	13:02:48:15	13:02:20:41
5	☆ lijinshu	5	77981	7:01:15:00 (-2)	7:00:43:28 (-1)	13:06:06:00	13:12:11:00	13:06:26:00
6	☆ chenxinmiao (陈欣妙)	4	33252	1:22:07:00	7:00:37:00	7:01:17:00		7:01:51:00 (-1)
7	☆ junyan007008	3	39948	7:01:07:00 (-1)	7:01:22:40 (-1)	13:12:19:00 (-7)		
8	☆ wyb1111 (王毅博)	2	19983	7:05:54:00 (-3)	6:14:09:00			
9	☆ chujinxuan (褚锦轩)	2	20430	7:00:39:00 (-4)	7:01:31:00 (-3)			

上周作业链接: https://vjudge.net/contest/805933

本周作业

https://vjudge.net/contest/807399 (课上讲了上周的 A ~ E 题, 课后作业是本周的 A ~ E 题)

课堂表现

这周的题目比起之前要稍微难一些, 同学们课上做题比较少, 课下要多补补这周的题。

课堂内容

CF1153D Serval and Rooted Tree

f[i]: 第 i 个数最少由几个叶子确定它的值

```
#include <bits/stdc++.h>

using namespace std;
```

```
const int maxn = 3e5 + 5;
vector<int> vec[maxn];
int w[maxn], f[maxn];
int cnt = 0;

void dfs(int u) {
  if (vec[u].empty()) { ++cnt; f[u] = 1; return; }

  int sum = 0, minn = 1e9;
  for (int i : vec[u]) {
    dfs(i);
    sum += f[i], minn = min(minn, f[i]);
  }

  f[u] = (w[u] ? minn : sum);
}

int main()
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i];
  for (int i = 2; i <= n; ++i) {
    int x; cin >> x; vec[x].push_back(i);
  }

  dfs(1);
  cout << cnt - f[1] + 1 << endl;
  return 0;
}
```

CF1286A Garland

f[i][j][0/1]: 考虑前 i 个数中, 一共有 j 个奇数, 最后以 奇数/偶数 结尾时, 最小的复杂度是多少

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 100 + 5;
int w[maxn];
int f[maxn][maxn][2];

int main()
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i];

  memset(f, 0x3f, sizeof(f));
  f[0][0][0] = f[0][0][1] = 0;
  for (int i = 1; i <= n; ++i) {
```

```
    for (int j = 0; j <= i; ++j) {
      if (w[i]==0 || w[i]%2 == 1) {
        if (j >= 1) f[i][j][1] = min(f[i-1][j-1][0]+1, f[i-1][j-1][1]);
      }
      if (w[i]==0 || w[i]%2 == 0) {
        f[i][j][0] = min(f[i-1][j][0], f[i-1][j][1]+1);
      }
    }
  }

  cout << min(f[n][(n+1)/2][0], f[n][(n+1)/2][1]) << endl;
  return 0;
}
```

CF1886C Decreasing String

可以用枚举或二分确定最后一段是删了几个字符, 然后用单调栈删掉这些数量的字符即可

```
#include <bits/stdc++.h>

using namespace std;

typedef long long LL;

LL get_sum(int l, int r) { return ((LL)l+r)*(r-l+1)/2; }

bool check(int n, int mid, LL pos) {
  LL sum = get_sum(n-mid+1, n);
  return pos > sum;
}

string get_str(string s, int k) {
  vector<char> vec;
  for (char i : s) {
    while (!vec.empty() && i<vec.back() && k) vec.pop_back(), --k;
    vec.push_back(i);
  }
  while (k) vec.pop_back(), --k;

  string res;
  for (char i : vec) res += i;
  return res;
}

void solve() {
  string s; LL pos; cin >> s >> pos;
  int n = s.size();

  int l = 0, r = n-1;
  while (l <= r) {
    int mid = (l + r) / 2;
```

```
    if (check(n, mid, pos)) l = mid+1;
    else r = mid-1;
  }

  LL sum = get_sum(n-r+1, n);
  pos -= sum;

  string s2 = get_str(s, r);

  cout << s2[pos-1];
}

int main()
{
  int T; cin >> T;
  while (T -- ) solve();
  return 0;
}
```

CF1883E Look Back

先求每个数跟前一个数的差距, 差几次 乘2 或者 除2 的操作, 然后从前往后做累加

```
#include <bits/stdc++.h>

using namespace std;

typedef long long LL;
const int maxn = 1e5 + 5;
int w[maxn], h[maxn];

int calc(int a, int b) {
  int res = 0;
  while (a*2 <= b) a *= 2, --res;
  while (a > b) b *= 2, ++res;
  return res;
}

void solve() {
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i];
  for (int i = 2; i <= n; ++i) h[i] = calc(w[i-1], w[i]);

  LL t = 0, res = 0;
  for (int i = 2; i <= n; ++i) {
    t += h[i];
    if (t < 0) t = 0;
    res += t;
  }

//  cout << "-------------------------- ";
```

```
  cout << res << endl;
}

int main()
{
  int T; cin >> T;
  while (T -- ) solve();
  return 0;
}
```

CF14E Camels

dp, f(i,j,k,x,y): 第i位是j, i-1位是k, 有x个高峰,y个低峰时 的方案数

```
#include <bits/stdc++.h>

using namespace std;

const int N = 20 + 5, M = 10 + 5;
int f[N][5][5][M][M];
// f(i,j,k,x,y): 第i位是j, i-1位是k, 有x个高峰,y个低峰时 的方案数

int main()
{
  int n, t; cin >> n >> t;
  for (int i = 1; i <= 4; ++i) {
    for (int j = 1; j <= 4; ++j) {
      if (i == j) continue;
      f[2][i][j][0][0] = 1;
    }
  }

  for (int i = 3; i <= n; ++i) {
    for (int j = 1; j <= 4; ++j) {
      for (int k = 1; k <= 4; ++k) {
        if (j == k) continue;
        for (int x = 0; x <= t; ++x) {
          for (int y = 0; y <= t-1; ++y) {
            // f[i][j][k][x][y] <- f[i-1][k][l][?][?]
            // l, k, j
            for (int l = 1; l <= 4; ++l) {
              if (k == l) continue;
              if (l>k && k<j) {
                if (y >= 1) f[i][j][k][x][y] += f[i-1][k][l][x][y-1];
              } else if (l<k && k>j) {
                if (x >= 1) f[i][j][k][x][y] += f[i-1][k][l][x-1][y];
              } else {
                f[i][j][k][x][y] += f[i-1][k][l][x][y];
              }
            }
          }
```

```
            }
          }
        }
      }

      int res = 0;
      for (int i = 1; i <= 4; ++i) {
        for (int j = 1; j <= 4; ++j) res += f[n][i][j][t][t-1];
      }
      cout << res << endl;
      return 0;
    }
```